

Project Charter – Canadian Household consumption Dashboard

Final Project Assignment – Project Definition

Group - Members

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Background

Canadian household spending patterns have evolved significantly since 2020 due to the COVID-19 pandemic, rising inflation, and shifts in monetary policy. Statistics Canada publishes detailed datasets on household income, expenditure, and consumption, segmented by province, income quintile, and household disposable income, primarily at quarterly or annual frequencies. In parallel, monthly Consumer Price Index (CPI) data and interest rate movements are released separately through Statistics Canada and the Bank of Canada. Although these sources are rich and reliable, they are distributed across multiple tables and platforms, making integrated economic analysis difficult.

Currently, users must manually navigate static tables or isolated interactive views to compare income, spending, inflation, and interest rate trends. For example, analyzing how inflation and interest rate changes affect household final consumption expenditure—particularly food spending, one of the most inflation-sensitive categories—requires reconciling datasets with different time ranges, frequencies, and dimensional structures. Food expenditure data, available annually with detailed category breakdowns from 2016 to 2023, is not natively linked to broader household expenditure or monthly inflation indicators within existing tools.

This BI initiative proposes a centralized Business Intelligence platform that integrates income, expenditure, food spending, inflation, and interest rate data into a unified dimensional model. By aligning quarterly, annual, and monthly datasets using conformed dimensions such as province, income quintile, time, CPI, and interest rate indicators, the platform will enable intuitive drill-downs, trend analysis, and cross-factor comparisons. The resulting dashboards will support deeper insight into how macroeconomic conditions influence household spending behavior across regions and income groups in Canada.

Goals

- **Unify disparate public datasets** by integrating Statistics Canada income, expenditure, and food spending tables with Bank of Canada CPI and interest rate data into a single, consistent BI data model.
- **Enable multi-dimensional analysis** of household spending trends by province, income quintile, disposable income level, and time period (2020–2025).
- **Provide focused drilldowns on food expenditure (2016–2023)** to analyze category-level sensitivity to inflation and income segmentation.
- **Overlay macroeconomic indicators** (monthly CPI and interest rate movements) as dimension tables to support correlation and impact analysis on household consumption patterns.
- **Improve accessibility and insight generation** through interactive dashboards that reduce reliance on manual table exploration and data reconciliation across platforms.

Scope

The project will deliver an **interactive analytical dashboard** that examines **Canadian household expense behavior** in relation to **inflation, household income, and interest rate changes**. The dashboard will integrate publicly available data from **Statistics Canada** and **Bank of Canada** sources, covering the period **2016 to 2026**.

The analysis will focus on the relationships between:

- Inflation trends (Consumer Price Index)
- Food spending patterns
- Household income and consumption levels
- Bank rate and mortgage rate changes

The dashboard will enable users to explore how changes in inflation and interest rates influence consumer spending behavior over time. Data will be cleaned, standardized, and modeled to ensure consistency across sources and time periods.

In Scope:

- Integration of the following datasets:
 - Inflation (CPI): Statistics Canada Table 18-10-0006-01
 - Food Spending: Statistics Canada Table 11-10-0125-01
 - Household Income: Statistics Canada Table 36-10-0663-01
 - Bank rate and mortgage rate data (March 2016–March 2026)
- Development of an interactive dashboard with:
 - Time-series visualizations
 - Trend and comparison analysis
 - Key indicators and summary insights
- Exploratory analysis of consumer expense behavior in response to:
 - Inflationary pressure
 - Income changes
 - Interest rate fluctuations
- Optional advanced component:
 - Development of a **predictive model** to forecast future Canadian household expense behavior based on historical trends and macroeconomic variables

Outcome Expectation

The project will result in a **data-driven dashboard** that provides clear and actionable insights into **Canadian household expense behavior** over the last decade. Users will be able to visually assess how inflation, income levels, and interest rate changes have influenced spending patterns, particularly in essential categories such as food.

Key outcomes include:

- A consolidated view of major economic indicators affecting Canadian consumers
- Improved understanding of the relationship between inflation, income, and expenses
- Identification of historical trends and structural shifts in consumer behavior
- Enhanced data literacy through intuitive visualizations and storytelling

If the forecasting component is implemented, the project will also deliver:

- A baseline predictive model capable of estimating **future expense behavior**
- Scenario-based insights illustrating how potential inflation or interest rate changes may impact household spending

Overall, the project will support **evidence-based economic analysis** and provide a strong foundation for further research, policy discussion, or advanced business intelligence applications.

Project Duration

The project will run from January 15 to April 14, covering two phases: project charter development and full analytics delivery. All milestones from data collection to final presentation will be completed within this structured 12-week timeline.

Key Stakeholders

Client & Sponsor	Penny, Daniel (Faculty) -Technology and Environment
Project manager	Tiwari, Yashaswi (student) - NSCC
Project team members	Danielle Aranha, Chisom Njoku, Chukwuma Okoro, Thi Kim Thanh Pham (students) - NSCC

Project Milestones

Milestone	Timeline (Week/Days)	Description / Deliverables
Initial Idea Exploration (Budget Buddy Data)	Dec 2025	Began with a personal “Budget Buddy” dataset to explore financial tracking and basic analysis concepts.
Research & Feasibility Discussion	Jan 15 - 22	Evaluated limitations of personal data (size, scope, insights) and discussed the need for a broader dataset.
Research and Discuss “How to Transition to Public Dataset”	Jan 22 - 30	Shifted to public datasets to enable deeper, more meaningful analysis.
Dataset Selection & Understanding	Feb 1 - 30	Selected the dataset and reviewed the structure, variables, and relevance to the problem statement.
Team Discussion & Charter Planning	Mar 1 - 15	Team discussed project direction, defined the problem, and divided responsibilities for charter preparation (objectives, scope, milestones, key stakeholders, budget, methodology, tools)
Project Charter Development (Group Work)	Mar 16 - 25	Collaboratively developed charter sections.

Phase 1 Submission (Charter)	Mar 26	Finalized and submitted one complete group project charter
Data cleaning and preparation	Mar 27 - 31	Cleaned dataset: handled missing values, removed duplicates, formatted fields
Data Modelling (Star Schema Design)	Apr 1 - 5	Designed data model (fact & dimension tables) to structure data efficiently for analysis
Exploratory Data Analysis (EDA)	Apr 1 - 5	Identified trends, patterns, and relationships using summary statistics and visualization
Data Processing & KPI Development	Apr 6	Create calculated fields, KPIs and metrics to support analysis
Visualization & Dashboard Development	Apr 6 - 12	Build a dashboard (Power BI) to present insights clearly and interactively
Insight Generation & Recommendations	Apr 6 - 12	Interpreted results and developed a data-driven solution to the problem
Presentation Preparation	Apr 6 - 12	Prepared slides
Phase 2 Submission & Presentation	Apr 14	Submitted final deliverables (data + visuals) and presented findings to the class

Project Budget

Expense Category	Description	Cost (Range)	Type
Data analysis tools	Use of analytics and visualization tools like Power BI, Excel through free versions	\$0	Non-Recurring
Data sources	Publicly available datasets from Statistics Canada and Bank of Canada	\$0	Non-Recurring
Development resources	Student time for data cleaning, modeling, analysis, dashboard development, and testing	\$0	Non-Recurring
Hosting & platform	Publishing and sharing dashboards using free or college provided platforms	\$0 - \$50	Recurring

Testing & validation	Validation of calculations, trend analysis, and visual accuracy	\$0	Non-Recurring
Documentation & presentation	Preparation of project documentation, analytical report, and presentation materials	\$0	Non-Recurring

Constraints, Assumptions, Risks and Dependencies

Constraints	• Dependence on public datasets: Dependence on publicly available datasets from Statistics Canada and Bank of Canada; delays or format changes may affect analysis. • Variation in dataset structures: Variation in dataset frequencies and structures (monthly, annual, weekly) requires additional standardization effort. • Tool limitations: Limited to free tools (Power BI, Excel); no budget for advanced analytics or cloud services. • Analytics-only scope: Analytical scope only (dashboard-focused); no full application development or external data inputs beyond public datasets.
Assumptions	• Public datasets remain accessible and up to date: It is assumed that all required Statistics Canada and Bank of Canada datasets, CPI, food spending, household income, and interest rate data will remain publicly accessible throughout the project timeline, with no interruptions in availability or changes that prevent integration. • Free analytical tools provided by NSCC are sufficient: The project assumes that free versions of Power BI, Excel, and institution provided platforms will be fully sufficient for data cleaning, modeling, and dashboard development and that no paid tools will be required
Risks and Dependencies	Risks: • Data Quality or Update Delays: There is a risk that publicly available datasets from Statistics Canada and Bank of Canada may have missing values, inconsistent formats, or delayed updates, which could affect data cleaning, modeling accuracy, and dashboard reliability • Timeline and Workload Pressures: Given the fixed academic schedule (Jan 05–Apr 14), any delays in early tasks such as dataset collection or cleaning may cascade and reduce the time available for analysis, visualization, and final presentation. This can impact the completeness of the final dashboard • Limited Technical Resources: The project depends on free tools and student expertise. Lack of advanced analytics tools or limited technical

experience could restrict the complexity of modeling, reduce performance, or introduce errors in analysis.

Dependencies:

- **Availability of Public Data Sources:** The project depends on continuous availability and accessibility of datasets from Statistics Canada (CPI, food spending, income) and Bank of Canada (interest rate). Any changes to access, structure, or availability could disrupt the workflow.
- **NSCC Provided Tools and Platforms:** The project depends on NSCC-provided or free tools (Excel, Power BI free version, college platforms) for analysis and dashboard development. Any technical issues, access limitations, or platform unavailability may impede progress.
- **Team Member Participation:** Tasks such as data cleaning, analysis, visualization, and documentation rely on active participation by all group members throughout the project timeline. Reduced availability or uneven workload distribution may affect deliverable quality

Approval Signatures

Name: **Daniel Penny**

For Business Intelligence and
Data Analytics program

(NSCC)

Project Client

Name: **Daniel Penny**

Faculty, Technology and
Environment,

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Project Sponsor

Name: **Yashaswi Tiwari**

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Student, NSCC

Project Manager

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